fluid sensing and midar can be done using satellites or does it have to be like planes and drones?

7:59 p.m.

Fluid Lensing and MiDAR can be done using satellites, and they do not have to rely solely on planes and drones. IEEE Xplore +1

While these breakthrough NASA technologies were originally developed and tested on low-altitude unmanned aerial vehicles (UAVs/drones) and aircraft, NASA explicitly designed them with **spaceborne platforms like CubeSats (small satellites)** in mind. NASA (.gov) +2

The transition from drones to satellites depends on how each technology handles the physics of light: NASA (.gov)

- **Fluid Lensing (via FluidCam):** This is a passive imaging technology that uses advanced algorithms to "undo" the distortion created by ocean surface waves, revealing sub-centimeter 3D images of things like coral reefs or submerged objects. NASA has already engineered a second-generation **FluidCam payload specifically to fit inside a 6U CubeSat satellite architecture** so it can map the world's shallow oceans from orbit. ResearchGate +1
- **MiDAR (Multispectral Imaging, Detection, and Active Reflectance):** Unlike standard cameras, MiDAR is an *active* sensor. It shines its own high-intensity, coded LED light patterns at a target to instantly calculate spectral signatures and shoot through murky water, even in total darkness. Because it requires a lot of power to shine light all the way down from space, scaling it to a full orbital satellite is harder than mounting it on a drone or an underwater ROV. However, NASA has successfully modeled its data reconstruction for future spaceborne

Ask anything

- How **MiDAR's active illumination** works differently from traditional satellite radar/Lidar
- Current real-world drone operations for mapping **coral reefs or submerged military munitions** University of Miami +4

is cubesats just limited to corals or can it be used for general bathymetric surveys etc

8:01 p.m.

CubeSats are absolutely not limited to mapping corals; they are actively and widely

Commercial CubeSat constellations—such as Planet Labs' **PlanetScope SuperDove fleet**—along with NASA's specialized payloads, are currently used across multiple marine sectors. ResearchGate +1

General Bathymetric Applications for CubeSats

Beyond ecology, CubeSat-based fluid sensing is deployed for critical marine operations:

- **Nautical Charting & Shoal Detection:** Keeping navigation charts updated in dynamic coastal zones. CubeSats can flag dangerous, newly formed sandbars or shoals that could ground commercial ships long before a hydrographic survey vessel is dispatched to the area. MDPI +3
- **Coastal Engineering & Infrastructure:** Monitoring sediment transport, beach erosion, and the structural integrity of shallow seafloors around ports, piers, seawalls, and offshore wind farms. USGS (.gov) +1
- **Disaster Management:** Pre- and post-storm mapping. Following major hurricanes, CubeSats provide rapid "revisit" imaging to see how the seafloor shifted, identifying underwater debris or storm-surge hazards blocking critical harbors. Ocean Science & Techn… +2
- **Military Hydrography:** Scouting nearshore depth profiles for naval amphibious operations and detecting submerged obstacles or munitions without putting personnel in harm's way.

The Trade-offs of CubeSat Bathymetry

While CubeSats provide a massive leap in how frequently we can scan the Earth's coastlines, they operate under distinct environmental trade-offs compared to heavy ships and aircraft:

Attribute	Traditional Methods (Sonar/Airborne LiDAR)	CubeSat-Derived Bathymetry

Revisit Frequency	**Poor:** Weeks to years; requires scheduling expensive aircraft or tracking routes by ship.	**Exceptional:** Constellations can scan the exact same coastal zone **daily or weekly**.
Cost	**High:** Thousands of dollars per square kilometer for specialized field deployment.	**Extremely Low:** Scalable data analytics over massive regional and global spans.

How It Works Globally

Rather than using active lasers (LiDAR), CubeSats primarily use **passive empirical and physics-based algorithms**. They measure the exact ratio of sunlight reflecting off the seabed through blue and green spectral bands. Because blue and green light penetrate water at predictable, mathematically bound rates, advanced computer models (including machine learning) transform those color variances directly into 3D depth profiles. Archive ouverte HAL +2

The introduction of fluid lensing onto orbital CubeSats helps remove the final major barrier for passive sensors: wave distortion. By stripping away wave artifact errors, a CubeSat can run a general bathymetric survey of a shallow harbor with the kind of clarity normally reserved for a drone flying just a few hundred feet overhead. NASA (.gov) +1

If you are planning or researching a bathymetric project, let me know:

- What is the **approximate depth** of your target zone?
- Is the water generally **clear or turbid/muddy**?
- Do you require **centimeter-level precision**, or is **meter-level tracking** sufficient?

I can help clarify which sensing platform or mathematical algorithm fits your constraints.

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